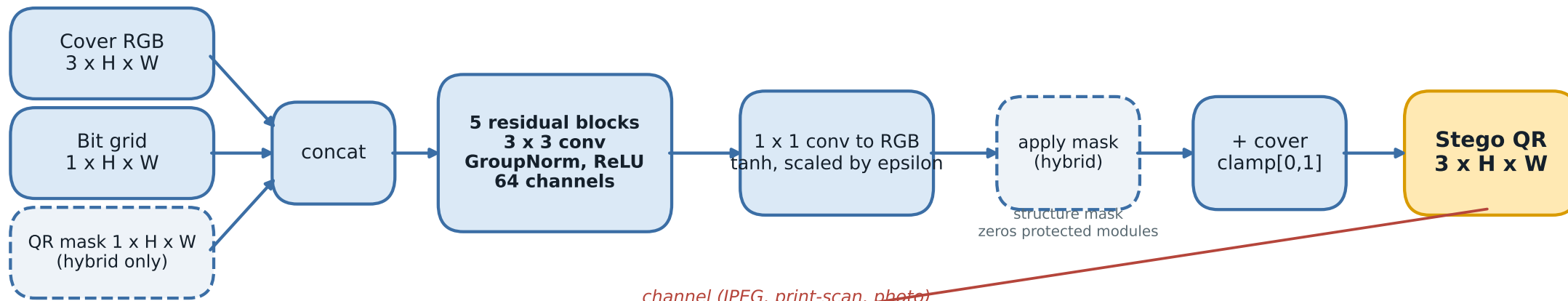
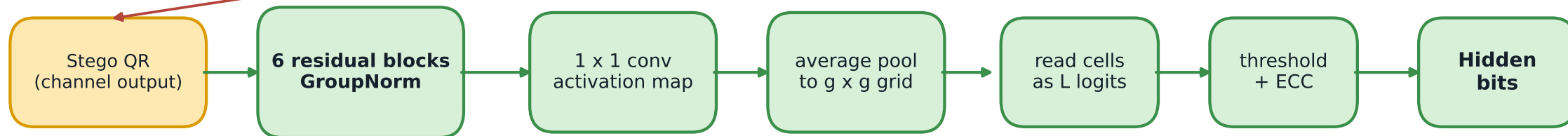


Encoder



Decoder



Spatial bit grid: each payload bit occupies one cell of a near-square grid upsampled to H x W; the decoder average-pools back to the grid and reads each cell.

Modes: segregated uses 3 independent channel networks; cross-channel uses joint RGB; hybrid adds a structural mask and data-cell placement.